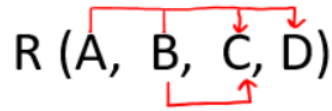


UNIT-2

24. Given a relation R(A, B, C, D) and Functional Dependency set $FD = \{ AB \rightarrow CD, B \rightarrow C \}$, determine whether the given R is in 2NF? If not convert it into 2 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attributes AB is not determined by any of the given FD, hence AB will be the integral part of the Candidate key, i.e. no matter what will be the candidate key, and how many will be the candidate key, but all will have W compulsory attribute.

Let us calculate the closure of AB

$AB^+ = ABCD$ (from the method we studied earlier)

Since the closure of AB contains all the attributes of R, hence **AB is Candidate Key**

From the definition of Candidate Key (**Candidate Key is a Super Key whose no proper subset is a Super key**)

Since all key will have AB as an integral part, and we have proved that AB is Candidate Key, Therefore, any superset of AB will be Super Key but not Candidate key.

Hence there will be only **one candidate key AB**

Definition of 2NF: No non-prime attribute should be partially dependent on Candidate Key

Since R has 4 attributes: - A, B, C, D, and Candidate Key is AB, Therefore, prime attributes (part of candidate key) are A and B while a non-prime attribute are C and D

R1(B, C)

Since the key is AB, and from FD $AB \rightarrow CD$, we can create $R2(A, B, C, D)$ but this will again have a problem of partial dependency $B \rightarrow C$, hence $R2(A, B, D)$.

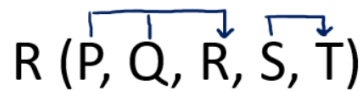
Finally, the decomposed table which is in 2NF

a) $R1(B, C)$

b) $R2(A, B, D)$

25. Given a relation R(P, Q, R, S, T) and Functional Dependency set $FD = \{ PQ \rightarrow R, S \rightarrow T \}$, determine whether the given R is in 2NF? If not convert it into 2 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attributes PQS is not determined by any of the given FD, hence PQS will be the integral part of the Candidate key, i.e., no matter what will be the candidate key, and how many will be the candidate key, but all will have PQS compulsory attribute.

Let us calculate the closure of PQS

$PQS^+ = PQSRT$ (from the method we studied earlier)

Since the closure of PQS contains all the attributes of R, hence **PQS is Candidate Key**

From the definition of Candidate Key (**Candidate Key is a Super Key whose no proper subset is a Super key**)

Since all key will have PQS as an integral part, and we have proved that PQS is Candidate Key.

Therefore, any superset of PQS will be Super Key but not Candidate key.

Hence there will be only **one candidate key PQS**

Definition of 2NF: No non-prime attribute should be partially dependent on Candidate Key.

Since R has 5 attributes: - P, Q, R, S, T and Candidate Key is PQS, Therefore, prime attributes (part of candidate key) are P, Q, and S while a non-prime attribute is R and T

a) FD: $PQ \rightarrow R$ does not satisfy the definition of 2NF, that non-prime attribute(R) is partially dependent on part of candidate key PQS.

b) FD: $S \rightarrow T$ does not satisfy the definition of 2NF, as a non-prime attribute(T) is partially dependent on candidate key PQS (i.e., key should not be broken at any cost).

Hence, FD $PQ \rightarrow R$ and $S \rightarrow T$, the above table R(P, Q, R, S, T) is not in 2NF

Convert the table R(P, Q, R, S, T) in 2NF:

Since due to FD: $PQ \rightarrow R$ and $S \rightarrow T$, our table was not in 2NF, let's decompose the table

R1(P, Q, R) (Now in table R1 FD: $PQ \rightarrow R$ is Full F D, hence R1 is in 2NF)

R2(S, T) (Now in table R2 FD: $S \rightarrow T$ is Full F D, hence R2 is in 2NF)

And create one table for the key, since the key is PQS.

R3(P, Q, S)

Finally, the decomposed tables which is in 2NF are:

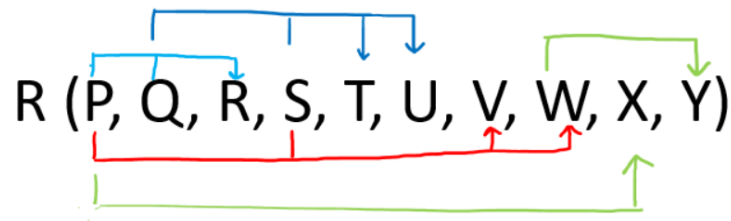
a) **R1(P, Q, R)**

b) **R2(S, T)**

c) **R3(P, Q, S)**

26. Given a relation R (P, Q, R, S, T, U, V, W, X, Y) and Functional Dependency set FD = {PQ → R, PS → VW, QS → TU, P → X, W → Y}, determine whether the given R is in 2NF? If not convert it into 2 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attributes PQS is not determined by any of the given FD, hence PQS will be the integral part of the Candidate key, i.e. no matter what will be the candidate key, and how many will be the candidate key, but all will have PQS compulsory attribute.

Let us calculate the closure of PQS

PQS⁺ = P Q S R T U V W X Y (from the closure method we studied earlier)

Since the closure of PQS contains all the attributes of R, hence **PQS is Candidate Key**

From the definition of Candidate Key(**Candidate Key is a Super Key whose no proper subset is a Super key**)

Since all key will have PQS as an integral part, and we have proved that PQS is Candidate Key, Therefore, any superset of PQS will be Super Key but not a Candidate key.

Hence there will be only **one candidate key PQS**

Definition of 2NF: No non-prime attribute should be partially dependent on Candidate Key

Since R has 10 attributes: - P, Q, R, S, T, U, V, W, X, Y, and Candidate Key is PQS calculated using FD = { PQ → R, PS → VW, QS → TU, P → X, W → Y }. Therefore, prime attribute(part of candidate key) are P, Q, and S while non-prime attribute are R, T, U, V, W, X and Y

- a. FD: **PQ → R** does not satisfy the definition of 2NF, that non-prime attribute(R) is partially dependent on part of candidate key PQS
- b. FD: **PS → VW** does not satisfy the definition of 2NF, that non-prime attribute(VW) is partially dependent on part of candidate key PQS
- c. FD: **QS → TU** does not satisfy the definition of 2NF, that non-prime attribute(TU) is partially dependent on part of candidate key PQS
- d. FD: **P → X** does not satisfy the definition of 2NF, that non-prime attribute(X) are partially dependent on part of candidate key PQS
- e. FD: **W → Y** does **not violate** the definition of 2NF, as the non-prime attribute(Y) is dependent on the non-prime attribute(W), which is not related to the definition of 2NF.

Hence because of FD: $PQ \rightarrow R$, $PS \rightarrow VW$, $QS \rightarrow TU$, $P \rightarrow X$ the above table $R(P, Q, R, S, T, U, V, W, X, Y)$ is not in 2NF

Convert the table $R(P, Q, R, S, T, U, V, W, X, Y)$ in 2NF:

Since due to FD: $PQ \rightarrow R$, $PS \rightarrow VW$, $QS \rightarrow TU$, $P \rightarrow X$ our table was not in 2NF, let's decompose the table

R1(P, Q, R) (Now in table R1 FD: $PQ \rightarrow R$ is Full F D, hence R1 is in 2NF)

R2(P, S, V, W) (Now in table R2 FD: $PS \rightarrow VW$ is Full F D, hence R2 is in 2NF)

R3(Q, S, T, U) (Now in table R3 FD: $QS \rightarrow TU$ is Full F D, hence R3 is in 2NF)

R4(P, X) (Now in table R4 FD: $P \rightarrow X$ is Full F D, hence R4 is in 2NF)

R5(W, Y) (Now in table R5 FD: $W \rightarrow Y$ is Full F D, hence R2 is in 2NF)

And create one table for the key, since the key is PQS.

R6(P, Q, S)

Finally, the decomposed tables which is in 2NF are:

R1(P, Q, R)

R2(P, S, V, W)

R3(Q, S, T, U)

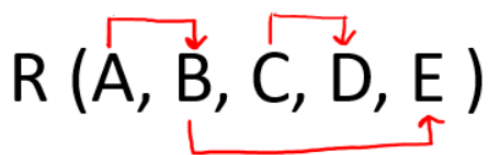
R4(P, X)

R5(W, Y)

R6(P, Q, S)

27. Given a relation $R(A, B, C, D, E)$ and Functional Dependency set $FD = \{A \rightarrow B, B \rightarrow E, C \rightarrow D\}$, determine whether the given R is in 2NF? If not convert it into 2 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attributes AC is not determined by any of the given FD, hence AC will be the integral part of the Candidate key, i.e. no matter what will be the candidate key, and how many will be the candidate key, but all will have W compulsory attribute.

Let us calculate the closure of AC

$AC^+ = ACBED$ (from the closure method we studied earlier)

Since the closure of AC contains all the attributes of R, hence **AC is Candidate Key**

From the definition of Candidate Key(**Candidate Key is a Super Key whose no proper subset is a Super key**)

Since all key will have AC as an integral part, and we have proved that AC is Candidate Key, Therefore, any superset of AC will be Super Key but not Candidate key.

Hence there will be only **one candidate key AC**

Definition of 2NF: No non-prime attribute should be partially dependent on Candidate Key

Since R has 5 attributes: - A, B, C, D, E and Candidate Key is AC, Therefore, prime attribute (part of candidate key) are A and C while the non-prime attribute are B D and E

- a. FD: **A → B** does not satisfy the definition of 2NF, as a non-prime attribute(B) is partially dependent on candidate key AC (i.e., key should not be broken at any cost).
- b. FD: **B → E** does **not violate** the definition of 2NF, as a non-prime attribute(E) is dependent on the non-prime attribute(B), which is not related to the definition of 2NF.
- c. FD: **C → D** does not satisfy the definition of 2NF, as a non-prime attribute(D) is partially dependent on candidate key AC (i.e., key should not be broken at any cost)

Hence because of FD A → B and C → D, the above table R(A, B, C, D, E) is not in 2NF

Convert the table R(A, B, C, D, E) in 2NF:

Since due to FD: A → B and C → D our table was not in 2NF, let's decompose the table

R1(A, B, E) (from FD: A → B and B → E and both are violating 2 NF definition)

R2(C, D) (Now in table R2 FD: C → D is Full F D, hence R2 is in 2NF)

And create one table for candidate key AC

R3 (A, C)

Finally, the decomposed tables which are in 2NF:

- a. **R1(A, B, E)**
- b. **R2(C, D)**
- c. **R3(A, C)**

Procedure: To verify that given relational schema R is in 2NF or NOT, If NOT then Convert it to 2NF:

STEP 1: Calculate the Candidate Key of given R by using an arrow diagram on R.

STEP 2: Verify each FD with **Definition of 2NF** (No non-prime attribute should be partially dependent on Candidate Key)

STEP 3: Make a set of FD which do not satisfy 2NF, i.e. all those FD which are partial.

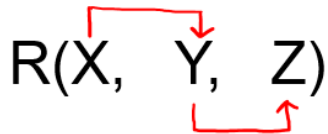
STEP 4: Convert the table R in 2NF by decomposing R such that each decomposition based on FD should satisfy the definition of 2NF:

STEP 5: Once the decomposition based on FD is completed, create a separate table of attributes in the Candidate key.

STEP 6: All the decomposed R obtained from STEP 4 and STEP 5 forms the required decomposition where each decomposition is in 2NF.

28. Given a relation R(X, Y, Z) and Functional Dependency set FD = { $X \rightarrow Y$ and $Y \rightarrow Z$ }, determine whether the given R is in 3NF? If not convert it into 3 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attribute X is not determined by any of the given FD, hence X will be the integral part of the Candidate key, i.e. no matter what will be the candidate key, and how many will be the candidate key, but all will have X compulsory attribute.

Let us calculate the closure of X

$X^+ = XYZ$ (from the closure method we studied earlier)

Since the closure of X contains all the attributes of R, hence **X is Candidate Key**

From the definition of Candidate Key (**Candidate Key is a Super Key whose no proper subset is a Super key**)

Since all key will have X as an integral part, and we have proved that X is Candidate Key, Therefore, any superset of X will be Super Key but not the Candidate key.

Hence there will be only **one candidate key X**

Definition of 3NF: A relational schema R is said to be in 3NF, First, it should be in 2NF and, no non-prime attribute should be transitively dependent on the Key of the table.

If $X \rightarrow Y$ and $Y \rightarrow Z$ exist then $X \rightarrow Z$ also exists which is a transitive dependency, and it should not hold.

Since R has 3 attributes: - X, Y, Z, and Candidate Key is X, Therefore, prime attribute (part of candidate key) is X while a non-prime attribute are Y and Z

Given FD are $X \rightarrow Y$ and $Y \rightarrow Z$

So, we can write $X \rightarrow Z$ (which is a transitive dependency)

In above **FD $X \rightarrow Z$** , a non-prime attribute(Z) is transitively depending on the key of the table(X) hence as per the definition of 3NF it is not in 3 NF, **because no non-prime attribute should be transitively dependent on the key of the table.**

Now check the above table is in 2 NF.

- FD: $X \rightarrow Y$ is in 2NF (as Key is not breaking and its Fully functional dependent)
- FD: $Y \rightarrow Z$ is also in 2NF(as it does not violate the definition of 2NF)

Hence above table R(X, Y, Z) is in 2NF but not in 3NF.

We can also prove the same from Definition 2: First, it should be in 2NF and if there exists a non-trivial dependency between two sets of attributes X and Y such that $X \rightarrow Y$ (i.e., Y is not a subset of X) then

- a. Either X is Super Key
- b. Or Y is a prime attribute.

Since we have just proved that above table R is in 2 NF. Let's check it for 3NF using definition 2.

- a. FD: $X \rightarrow Y$ is in 3NF (as X is a super Key)
- b. FD: $Y \rightarrow Z$ is not in 3NF (as neither Y is Key nor Z is a prime attribute)

Hence because of $Y \rightarrow Z$ using definition 2 of 3NF, we can say that above table R is not in 3NF.

Convert the table R(X, Y, Z) into 3NF:

Since due to FD: $Y \rightarrow Z$, our table was not in 3NF, let's decompose the table

FD: $Y \rightarrow Z$ was creating issue, hence one table R1(Y, Z)

Create one Table for key X, R2(X, Y), since $X \rightarrow Y$

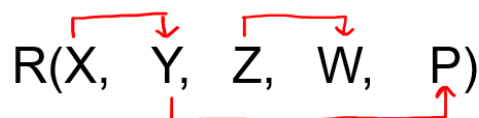
Hence decomposed tables which are in 3NF are:

R1(X, Y)

R2(Y, Z)

29. Given a relation R(X, Y, Z, W, P) and Functional Dependency set $FD = \{ X \rightarrow Y, Y \rightarrow P, \text{ and } Z \rightarrow W\}$, determine whether the given R is in 3NF? If not convert it into 3 NF.

Solution: Let us construct an arrow diagram on R using FD to calculate the candidate key.



From above arrow diagram on R, we can see that an attributes XZ is not determined by any of the given FD, hence XZ will be the integral part of the Candidate key, i.e. no matter what will be the candidate key, and how many will be the candidate key, but all will have XZ compulsory attribute.

Let us calculate the closure of XZ

$XZ^+ = XZYPW$ (from the closure method that we studied earlier)

Since the closure of XZ contains all the attributes of R, hence **XZ is Candidate Key**

From the definition of Candidate Key (**Candidate Key is a Super Key whose no proper subset is a Super key**).

Since all key will have XZ as an integral part, and we have proved that XZ is Candidate Key, Therefore, any superset of XZ will be Super Key but not the Candidate key.

Hence there will be only **one candidate key XZ**

Definition of 3NF: First it should be in 2NF and if there exists a non-trivial dependency between two sets of attributes X and Y such that $X \rightarrow Y$ (i.e., Y is not a subset of X) then

- a. Either X is Super Key
- b. Or Y is a prime attribute.

Since R has 5 attributes: - X, Y, Z, W, P and Candidate Key is XZ, Therefore, prime attribute (part of candidate key) are X and Z while a non-prime attribute are Y, W, and P

Given FD are $X \rightarrow Y$, $Y \rightarrow P$, and $Z \rightarrow W$ and Super Key / Candidate Key is XZ

- a. FD: $X \rightarrow Y$ does not satisfy the definition of 3NF, that neither X is Super Key nor Y is a prime attribute.
 - b. FD: $Y \rightarrow P$ does not satisfy the definition of 3NF, that neither Y is Super Key nor P is a prime attribute.
 - c. FD: $Z \rightarrow W$ satisfies the definition of 3NF, that neither Z is Super Key nor W is a prime attribute.

Convert the table R(X, Y, Z, W, P) into 3NF:

Since all the FD = { $X \rightarrow Y$, $Y \rightarrow P$, and $Z \rightarrow W$ } were not in 3NF, let us convert R in 3NF

R1(X, Y) {Using FD $X \rightarrow Y$ }

R2(Y, P) {Using FD $Y \rightarrow P$ }

R3(Z, W) {Using FD $Z \rightarrow W$ }

And create one table for Candidate Key XZ

R4(X, Z) { Using Candidate Key XZ }

All the decomposed tables R1, R2, R3, and R4 are in 2NF(as there is no partial dependency) as well as in 3NF.

Hence decomposed tables are:

R1(X, Y), R2(Y, P), R3(Z, W), and R4(X, Z)

16. Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$. What is the key for R ? Decompose R into 2NF and then 3NF relations

Solution:

$A \rightarrow DE$ (given) $\Rightarrow A \rightarrow D$ and $A \rightarrow E$

Since $A \rightarrow D$ and $D \rightarrow IJ$ (given) $\Rightarrow A \rightarrow IJ$

Using the union rule $A \rightarrow ADEIJ$, thus $AB \rightarrow ABDEIJ$ (augmentation)

Also $AB \rightarrow C$ (given) $\Rightarrow AB \rightarrow ABCDEIJ$.

Since $B \rightarrow F$ (given) and $F \rightarrow GH$ (given), $B \rightarrow GH$ (transitivity)

Thus $AB \rightarrow AGH$ holds. Also $AB \rightarrow AF$ holds from $B \rightarrow F$ (given)

Finally, using the union rule $AB \rightarrow ABCDEFGHIJ$.

So AB is a key. This can also be determined by calculating AB^+ with respect to the set F .

2NF

R1 (A, B, C)

R2 (A, D, E, I, J)

R3 (B, F, G, H)

3NF

R1 (A, B, C)

R2.1 (A, D, E) R2.2 (D, I, J)

R3.1 (B, F) R3.2 (F, G, H)

20. Prove that any relation schema with two attributes is in BCNF.

For any relation to be in BCNF, the following must hold for any functional dependency $X \rightarrow Y$:

Either $X \rightarrow Y$ is a Trivial Functional Dependency (i.e. $Y \subseteq X$) OR X is a superkey for schema R .

A relation with only two attributes is always in BCNF.

For Example, Assume relation $R = \{A, B\}$ with two attributes. The only possible Non-trivial FD's are $\{A\} \rightarrow \{B\}$ and $\{B\} \rightarrow \{A\}$.

So, there are four possible cases:

1. No Non-trivial FD's hold in R . i.e. $\{Cand. Key = AB\}$, Since it is an all key relation it's always in *BCNF*.
2. Only Non-trivial FD $A \rightarrow B$ holds. In this case, $\{Cand. Key = A\}$, and relation satisfies BCNF.
3. Only Non-trivial FD $B \rightarrow A$ holds. In this case, $\{Cand. Key = B\}$, and relation satisfies BCNF.
4. Both $A \rightarrow B$ and $B \rightarrow A$ holds. In this case, $\{Cand. Keys = A, B\}$, and relation satisfies BCNF.

Hence, every Binary Relation (A relation with two attributes) is always in BCNF!